

Web Engineering — Lab 02 Report

HTML Deep Dive — Semantic Elements, Forms & Tables

Student Name: Muhammad Tayyab	Course: Web Engineering (5th Semester)
CMS ID: 023-24-0118	Lab: Week 02: HTML Deep Dive

1. Lab Summary

In Week 02, we learned how to use semantic HTML5 tags instead of just using `<div>` tags everywhere. We practiced building an accessible profile page with internal jump links, adding images with `<figure>` and `<figcaption>`, creating forms with proper `<label>` and `<input>` types, building data tables, and passing the W3C validator.

2. Semantic HTML Elements Used

- `<header>`: Top banner with my name, tagline, and navigation.
- `<nav>`: Contains navigation links (`#about`, `#skills`, `#contact`).
- `<main>`: The main content of the page (used only once per page).
- `<article>`: Self-contained section for the About Me card.
- `<section>`: Thematic sections for Skills and Contact form.
- `<aside>`: Sidebar / extra info about lab timings.
- `<footer>`: Footer with copyright and social links.

3. In-Class Lab Deliverable (Week 02/profile.html)

Week 02/profile.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Profile page of Muhammad Tayyab" />
  <title>Muhammad Tayyab - Profile</title>
</head>
<body>
  <!-- Header -->
  <header>
    <h1>Muhammad Tayyab</h1>
    <p>BS Computer Science / Software Engineering Student (CMS ID: 023-24-0118)</p>
    <nav>

      <ul>
        <li><a href="#about">About</a></li>
        <li><a href="#skills">Skills</a></li>
        <li><a href="#contact">Contact</a></li>
        <li><a href="contact.html">Contact Page</a></li>
      </ul>
    </nav>
  </header>

  <!-- Main Content -->
  <main>
    <!-- About Section -->
    <article id="about" class="card">
      <h2>About Me</h2>

      <p>Hi, I am Muhammad Tayyab. I am learning web engineering this semester.</p>
      <p class="muted">Currently learning HTML5, CSS3, and JavaScript basics.</p>
      <figure>
        
        <figcaption>My coding workspace setup.</figcaption>
      </figure>
    </article>

    <!-- Skills Table Section -->
    <section id="skills" class="card">
      <h2>My Tools & Skills</h2>
      <table>
        <caption>Tools and Comfort Level</caption>
        <thead>

          <tr>
            <th scope="col">Tool</th>
            <th scope="col">Comfort</th>
          </tr>
        </thead>
        <tbody>
          <tr>
            <th scope="row">HTML5</th>
            <td>Strong</td>
          </tr>
          <tr>
            <th scope="row">CSS3</th>
            <td>Comfortable</td>
          </tr>
        </tbody>
      </table>
    </section>
  </main>
</body>
</html>
```

```

        <tr>
            <th scope="row">Git & GitHub</th>
            <td>Comfortable</td>
        </tr>
        <tr>
            <th scope="row">JavaScript</th>
            <td>Beginner</td>
        </tr>
    </tbody>
</table>
</section>

<!-- Contact Form Section -->
<section id="contact" class="card">

    <h2>Contact Me</h2>
    <form action="#" method="post">
        <div>
            <label for="name">Full Name *</label>
            <input type="text" id="name" name="name" required />
        </div>
        <div>
            <label for="email">Email *</label>
            <input type="email" id="email" name="email" required placeholder="name@example.com" />
        </div>
        <div>
            <label for="phone">Phone</label>
            <input type="tel" id="phone" name="phone" />
        </div>

        <fieldset>
            <legend>Preferred Track</legend>
            <label><input type="radio" name="track" value="frontend" checked /> Frontend</label>
            <label><input type="radio" name="track" value="backend" /> Backend</label>
        </fieldset>
        <div>
            <label for="topic">Topic</label>
            <select id="topic" name="topic">
                <option value="">Choose one</option>
                <option value="help">Help</option>
                <option value="feedback">Feedback</option>
            </select>
        </div>
        <div>

            <label for="msg">Message</label>
            <textarea id="msg" name="message" rows="4"></textarea>
        </div>
        <div>
            <label><input type="checkbox" name="terms" required /> I accept the terms</label>
        </div>
        <button type="submit">Send</button>
    </form>
</section>
</main>

<aside>
    <h3>Lab Hours</h3>
    <p>Monday & Wednesday: 2 PM - 4 PM</p>

```

```

</aside>

<!-- Footer -->
<footer>
  <p>&copy; 2026 Muhammad Tayyab (023-24-0118)</p>
  <p>
    <a href="https://github.com" target="_blank" rel="noopener noreferrer">GitHub</a> |
    <a href="https://w3.org" target="_blank" rel="noopener noreferrer">W3C</a>
  </p>
</footer>
</body>
</html>

```

4. Form Testing (Activity D)

- **Label Click Test:** Clicking on the text label 'Full Name *' immediately placed the cursor inside the text box because `<label for="name">` matches `<input id="name">`. When I removed the `for` attribute, clicking the label did nothing.
- **Required Validation:** When I tried to submit the form without filling the required name and email fields, the browser blocked submission with a tooltip saying 'Please fill out this field'.
- **External Link Security:** Added `rel="noopener noreferrer"` to links with `target="_blank"` for security.

5. Homework Task 2: Rewriting Bad HTML (`homework-rewrite.html`)

```

homework-rewrite.html

<!-- Refactored with semantic HTML -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>My Blog</title>
</head>
<body>
  <header>
    <h1>My Blog</h1>
    <nav>
      <a href="/">Home</a>
    </nav>
  </header>
  <main>
    <article class="post">
      <header>
        <h2>First Post</h2>
        <time datetime="2026-08-09">Aug 9 2026</time>
      </header>
      <p>Hello world</p>
    </article>
  </main>
</body>
</html>

```

6. Homework Task 3 & 4 (Second Page & Reflection)

- **Task 3:** Created `contact.html` with a form and linked `profile.html` and `contact.html` to each other in both nav bars and footers.
- **Task 4 (Reflection):**
 1. I was going to use a `<div>` for the about section, but I used `<article>` instead because it represents a standalone piece of content.
 2. For extra sidebar notes, I used `<aside>` instead of a `div`.
 3. Adding `for` and `id` to all labels made the form much easier to click on.
 4. I used `<table>` only for the skills data with `scope="col"` and `scope="row"` as taught.
 5. I tested the page on `validator.w3.org` and got a green pass with 0 errors.